Mixed mathematical report

Section 1

한글 설명과 mathematical notation in one report.

1.1 Quadratic roots — before $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$
	Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

1.2 Continued fraction — before $1 + \cfrac{1}{2 + \cfrac{1}{3}}$ after *formatted explanation*.

1.3 Binomial coefficient — before $\binom{n}{k}$ after *formatted explanation*.

1.4 Nested roots — before $\sqrt[3]{x + \sqrt{y + \sqrt{z}}}$ after *formatted explanation*.

1.5 Indexed tensor — before $T^{k\,l}_{i\,j} = a_{i}\, b^{k}$ after *formatted explanation*.

1.6 Series with limits — before $\sum_{k=1}^{n} \frac{1}{k^2}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$

Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

1.7 Improper integral — before $\int_0^{\infty} e^{-x^2}\,dx$ after *formatted explanation*.

1.8 Partial derivatives — before $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ after *formatted explanation*.

1.9 Matrix with expressions — before $\begin{bmatrix} \frac{1}{2} & \sqrt{2} \\ a_i & b^2 \end{bmatrix}$ after *formatted explanation*.

1.10 Piecewise values — before $\begin{cases} x^2 & x \geq 0 \\ -x & x<0 \end{cases}$ after *formatted explanation*.

1.11 Vector and hat accents — before $\vec{v}+\hat{x}$ after *formatted explanation*.

Context	**Equation and formatting**
Merged cell	**Ratio**: $\frac{a}{b}$
	Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

1.12 Overline and underline — before $\overline{AB}+\underline{x}$ after *formatted explanation*.

1.13 Limits of a ratio — before $\lim_{x \to 0} \frac{\sin x}{x}=1$ after *formatted explanation*.

1.14 Probability — before $P\left(A \mid B\right)=\frac{P\left(A \cap B\right)}{P\left(B\right)}$ after *formatted explanation*.

1.15 Mathematical alphabets — before $\mathbb{R} \supset \boldsymbol{v} \in \mathcal{F}+\alpha$ after *formatted explanation*.

1.16 Semantic annotation — before $\frac{\alpha}{\beta}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

1.17 Phantom alignment space — before $x+y$ after *formatted explanation*.

1.18 Custom separators — before $\left\langle x \,;\, y \,\middle|\, z \,\middle|\, w \right\rangle$ after *formatted explanation*.

1.19 Bevelled fraction — before $\frac{a+b}{c}$ after *formatted explanation*.

1.20 Text and quoted string — before $f\left(\ \mathit{input}\ \right)=\text{output value}$ after *formatted explanation*.

End of section.

Section 2

한글 설명과 mathematical notation in one report.

2.1 Quadratic roots — before $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link Caption.

2.2 Continued fraction — before $\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3}}}$ after *formatted explanation*.

2.3 Binomial coefficient — before $\binom{n}{k}$ after *formatted explanation*.

2.4 Nested roots — before $\sqrt[3]{x+\sqrt{y+\sqrt{z}}}$ after *formatted explanation*.

2.5 Indexed tensor — before ${T^{k\,l}_{i\,j}} = a_{i}\,b^{k}$ after *formatted explanation*.

2.6 Series with limits — before $\sum_{k=1}^{n} \frac{1}{k^2}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link  Caption.

2.7 Improper integral — before $\int_0^\infty e^{-x^2}\,dx$ after *formatted explanation*.

2.8 Partial derivatives — before $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ after *formatted explanation*.

2.9 Matrix with expressions — before $\begin{bmatrix} \frac{1}{2} & \sqrt{2} \\ a_i & b^2 \end{bmatrix}$ after *formatted explanation*.

2.10 Piecewise values — before $\begin{cases} x^2 & x \geq 0 \\ -x & x<0 \end{cases}$ after *formatted explanation*.

2.11 Vector and hat accents — before $\vec{v}+\hat{x}$ after *formatted explanation*.

Context	**Equation and formatting**
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link  Caption.

2.12 Overline and underline — before $\overline{AB}+\underline{x}$ after *formatted explanation*.

2.13 Limits of a ratio — before $\lim_{x \to 0} \frac{\sin x}{x}=1$ after *formatted explanation*.

2.14 Probability — before $P(A \mid B)=\frac{P(A \cap B)}{P(B)}$ after *formatted explanation*.

2.15 Mathematical alphabets — before $\mathbb{R} \supset \boldsymbol{v} \in \mathcal{F} + \alpha$ after *formatted explanation*.

2.16 Semantic annotation — before $\frac{\alpha}{\beta}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

2.17 Phantom alignment space — before $x + y$ after *formatted explanation*.

2.18 Custom separators — before $\langle x \,;\, y \mid z \mid w \rangle$ after *formatted explanation*.

2.19 Bevelled fraction — before $a + b \,/\, c$ after *formatted explanation*.

2.20 Text and quoted string — before $f\left(\textit{input} \right) = \text{output value}$ after *formatted explanation*.

End of section.

Section 3

한글 설명과 mathematical notation in one report.

3.1 Quadratic roots — before $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

3.2 Continued fraction — before $\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3}}}$ after *formatted explanation*.

3.3 Binomial coefficient — before $\binom{n}{k}$ after *formatted explanation*.

3.4 Nested roots — before $\sqrt[3]{x+\sqrt{y+\sqrt{z}}}$ after *formatted explanation*.

3.5 Indexed tensor — before $T^{k\,l}_{i\,j} = a_{\,i}\, b^{\,k}$ after *formatted explanation*.

3.6 Series with limits — before $\sum_{k=1}^{n} \frac{1}{k^2}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link  Caption.

3.7 Improper integral — before $\int_0^\infty e^{-x^2}\,dx$ after *formatted explanation*.

3.8 Partial derivatives — before $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ after *formatted explanation*.

3.9 Matrix with expressions — before $\begin{bmatrix} \frac{1}{2} & \sqrt{2} \\ a_i & b^2 \end{bmatrix}$ after *formatted explanation*.

3.10 Piecewise values — before $\begin{cases} x^2 & x \geq 0 \\ -x & x<0 \end{cases}$ after *formatted explanation*.

3.11 Vector and hat accents — before $\vec{v}+\hat{x}$ after *formatted explanation*.

Context	**Equation and formatting**
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link  Caption.

3.12 Overline and underline — before $\overline{AB}+\underline{x}$ after *formatted explanation*.

3.13 Limits of a ratio — before $\lim_{x \to 0} \frac{\sin x}{x}=1$ after *formatted explanation*.

3.14 Probability — before $P(A \mid B)=\frac{P(A \cap B)}{P(B)}$ after *formatted explanation*.

3.15 Mathematical alphabets — before $\mathbb{R} \supset \boldsymbol{v} \in \mathcal{F} + \alpha$ after *formatted explanation*.

3.16 Semantic annotation — before $\frac{\alpha}{\beta}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

3.17 Phantom alignment space — before $x + y$ after *formatted explanation*.

3.18 Custom separators — before $\langle x \,;\, y \mid z \mid w \rangle$ after *formatted explanation*.

3.19 Bevelled fraction — before $a + b \,/\, c$ after *formatted explanation*.

3.20 Text and quoted string — before $f\left(\mathit{input} \right) = \text{output value}$ after *formatted explanation*.

End of section.

Section 4

한글 설명과 mathematical notation in one report.

4.1 Quadratic roots — before $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

4.2 Continued fraction — before $\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3}}}$ after *formatted explanation*.

4.3 Binomial coefficient — before $\binom{n}{k}$ after *formatted explanation*.

4.4 Nested roots — before $\sqrt[3]{x+\sqrt{y+\sqrt{z}}}$ after *formatted explanation*.

4.5 Indexed tensor — before $T^{k\,l}_{i\,j} = a_{\,i}\, b^{\,k}$ after *formatted explanation*.

4.6 Series with limits — before $\sum_{k=1}^{n} \frac{1}{k^2}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link  Caption.

4.7 Improper integral — before $\int_0^\infty e^{-x^2}\,dx$ after *formatted explanation*.

4.8 Partial derivatives — before $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ after *formatted explanation*.

4.9 Matrix with expressions — before $\begin{bmatrix} \frac{1}{2} & \sqrt{2} \\ a_i & b^2 \end{bmatrix}$ after *formatted explanation*.

4.10 Piecewise values — before $\begin{cases} x^2 & x \geq 0 \\ -x & x<0 \end{cases}$ after *formatted explanation*.

4.11 Vector and hat accents — before $\vec{v}+\hat{x}$ after *formatted explanation*.

Context	**Equation and formatting**
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link  Caption.

4.12 Overline and underline — before $\overline{AB}+\underline{x}$ after *formatted explanation*.

4.13 Limits of a ratio — before $\lim_{x \to 0} \frac{\sin x}{x}=1$ after *formatted explanation*.

4.14 Probability — before $P(A \mid B)=\frac{P(A \cap B)}{P(B)}$ after *formatted explanation*.

4.15 Mathematical alphabets — before $\mathbb{R} \supset \boldsymbol{v} \in \mathcal{F} + \alpha$ after *formatted explanation*.

4.16 Semantic annotation — before $\frac{\alpha}{\beta}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

4.17 Phantom alignment space — before $x + y$ after *formatted explanation*.

4.18 Custom separators — before $\langle x \,;\, y \mid z \mid w \rangle$ after *formatted explanation*.

4.19 Bevelled fraction — before $a + b \,/\, c$ after *formatted explanation*.

4.20 Text and quoted string — before $f\left(\mathit{input} \right) = \text{output value}$ after *formatted explanation*.

End of section.

Section 5

한글 설명과 mathematical notation in one report.

5.1 Quadratic roots — before $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

5.2 Continued fraction — before $\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3}}}$ after *formatted explanation*.

5.3 Binomial coefficient — before $\binom{n}{k}$ after *formatted explanation*.

5.4 Nested roots — before $\sqrt[3]{x+\sqrt{y+\sqrt{z}}}$ after *formatted explanation*.

5.5 Indexed tensor — before $T^{k\,l}_{i\,j} = a_{\,i}\, b^{\,k}$ after *formatted explanation*.

5.6 Series with limits — before $\sum_{k=1}^{n} \frac{1}{k^2}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link Caption.

5.7 Improper integral — before $\int_0^{\infty} e^{-x^2}\,dx$ after *formatted explanation*.

5.8 Partial derivatives — before $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ after *formatted explanation*.

5.9 Matrix with expressions — before $\begin{bmatrix} \frac{1}{2} & \sqrt{2} \\ a_i & b^2 \end{bmatrix}$ after *formatted explanation*.

5.10 Piecewise values — before $\begin{cases} x^2 & x \geq 0 \\ -x & x<0 \end{cases}$ after *formatted explanation*.

5.11 Vector and hat accents — before $\vec{v}+\hat{x}$ after *formatted explanation*.

Context	**Equation and formatting**
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.

4. Keep **bold** and *italic* text.

Reference link Caption.

5.12 Overline and underline — before $\overline{AB}+\underline{x}$ after *formatted explanation*.

5.13 Limits of a ratio — before $\lim_{x \to 0} \frac{\sin x}{x}=1$ after *formatted explanation*.

5.14 Probability — before $P(A \mid B)=\frac{P(A \cap B)}{P(B)}$ after *formatted explanation*.

5.15 Mathematical alphabets — before $\mathbb{R} \supset \boldsymbol{v} \in \mathcal{F} + \alpha$ after *formatted explanation*.

5.16 Semantic annotation — before $\frac{\alpha}{\beta}$ after *formatted explanation*.

Context	Equation and formatting
Merged cell	**Ratio**: $\frac{a}{b}$ Nested table

3. Check the numerator and denominator.
4. Keep **bold** and *italic* text.

Reference link  Caption.

5.17 Phantom alignment space — before $x + y$ after *formatted explanation*.

5.18 Custom separators — before $\langle x \,;\, y \mid z \mid w \rangle$ after *formatted explanation*.

5.19 Bevelled fraction — before $a + b \,/\, c$ after *formatted explanation*.

5.20 Text and quoted string — before $f\left(\textit{input} \right) = \text{output value}$ after *formatted explanation*.